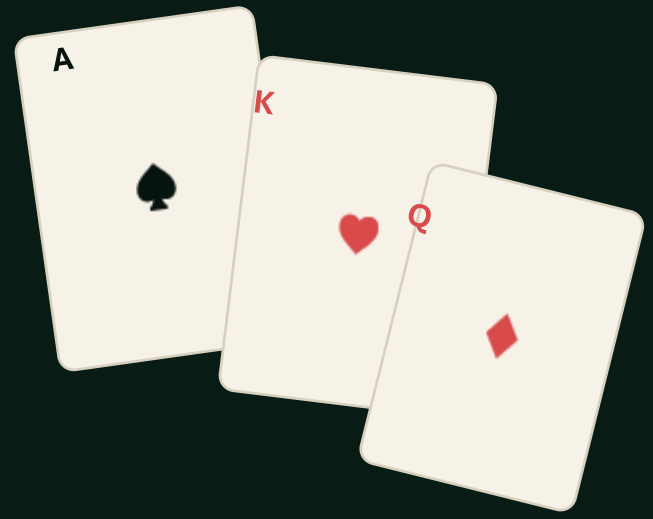
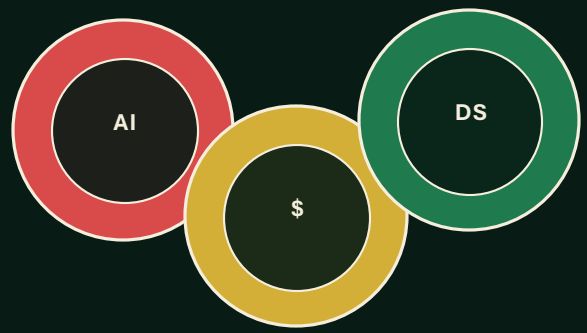
Where Should We Build Our Data Science Team?

Location strategy for an AI startup building casino & gaming technology



OUR STARTUP REALITY

We need strong data science talent—but we cannot outbid the biggest tech employers. Our advantage starts with choosing the right market.



Our Business Question

WHICH U.S. LOCATION SHOULD OUR AI GAMING STARTUP ESTABLISH ITS DATA SCIENCE TEAM IN BASED ON LOCAL COMPENSATION LEVELS, JOB-POSTING ACTIVITY, AND EMPLOYER COMPETITION?

LOWER COST

ACTIVE JOB MARKET

MANAGEABLE COMPETITION

What We’re Working With

THE DATASET

OUR FOCUS

WHY CLEANING MATTERS

Decision lens: Can this market give us enough talent access without forcing us into big-tech salary competition?

From Raw Postings to Decision-Ready Data

We cleaned for consistency first—then analyzed the market.



Audit

Review missing values, duplicates, formats, and relevant fields.

Clean

Standardize missing values and clean unusable formatting.

Standardize

Prepare salary, location, company, and job-title fields.

Validate

Check salary ranges, geographic categories, missing values, and row counts.

Analyze

Compare recruiting markets using cleaned data.

What Data Matters to Our Decision?

Our analysis depends on reliable compensation, location, and employer data.

COMPENSATION

Raw problem:
Salary estimates were stored as text ranges with symbols and labels, preventing direct numerical comparison.

LOCATION

Raw problem:
Location combined city and state, with state-only, remote, nationwide, and irregular geographic records.

EMPLOYERS

Raw problem:
Company names contained appended Glassdoor ratings that could interfere with identifying unique employers.

Why this matters: **These three fields directly determine how we measure affordability, market activity, and employer competition.**

Before vs. After

Real examples from our dataset showing how raw fields were transformed for analysis.

BEFORE

	Salary Estimate	Location	Company Name
16	$137K-$171K (Glassdoor est.)	Remote	Upside Business Travel\n4.1
93	$79K-$131K (Glassdoor est.)	Utah	Merrick Bank\n3.6
534	$66K-$112K (Glassdoor est.)	Patuxent, Anne Arundel, MD	MILVETS Systems Technology, Inc.\n3.4

AFTER

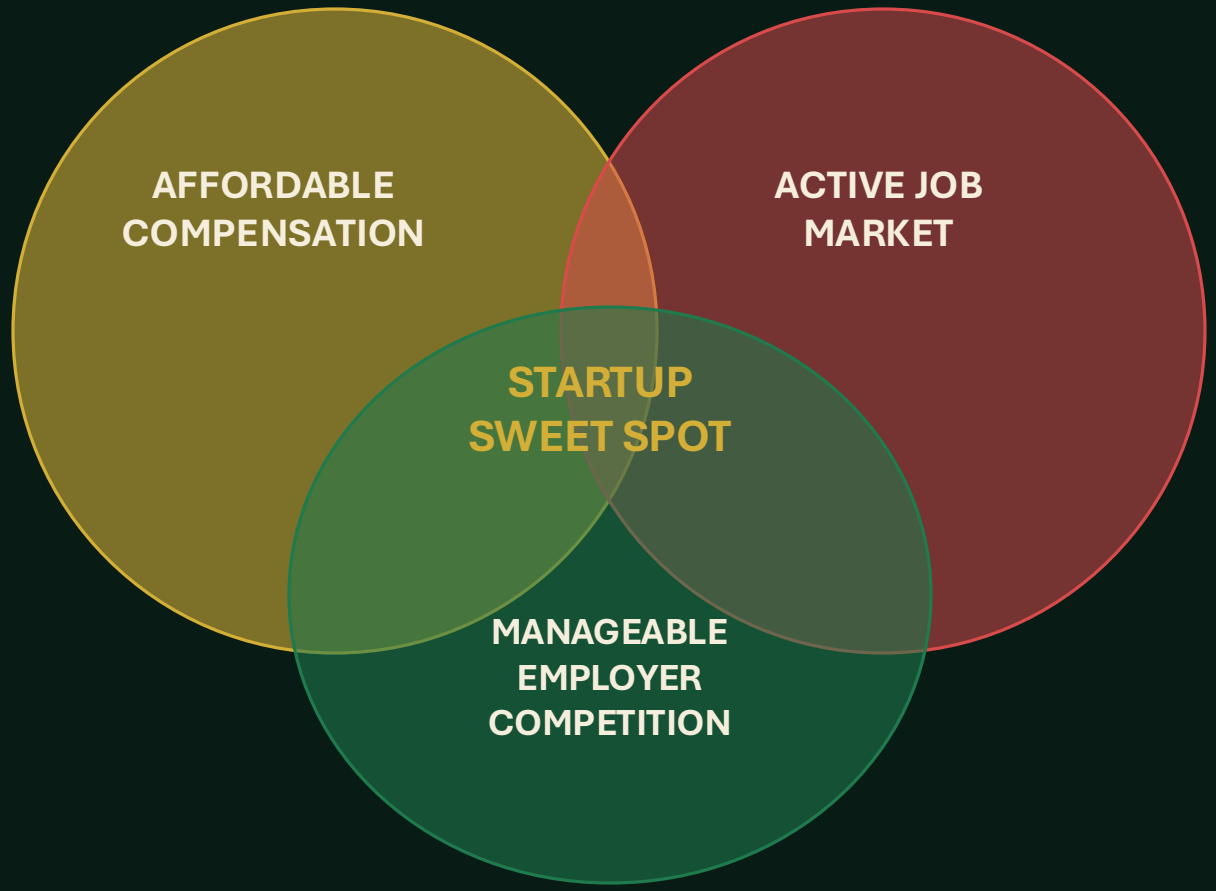
	min_salary	max_salary	avg_salary	city	state	Company Name
16	137000.0	171000.0	154000.0	NaN	NaN	Upside Business Travel
93	79000.0	131000.0	105000.0	NaN	UT	Merrick Bank
534	66000.0	112000.0	89000.0	Patuxent	MD	MILVETS Systems Technology, Inc.

672 total postings • 0 duplicates • Geographic records validated for location-based analysis

TOPIC FOCUS

Finding the Startup “Sweet Spot”

Our ideal market balances cost, activity, and competition.



WHAT WE’LL COMPARE

- Average compensation by state
- Job-posting activity by state
- Unique employers by state

The strongest location isn't simply the cheapest—it balances all three factors.

TOPIC FOCUS

Why These Factors Matter For Our Startup

LIMITED BUDGET

As a smaller startup, we need to recruit without competing at the highest salary levels.

ACTIVE MARKET

Posting activity helps identify states with an established market for data-science employment.

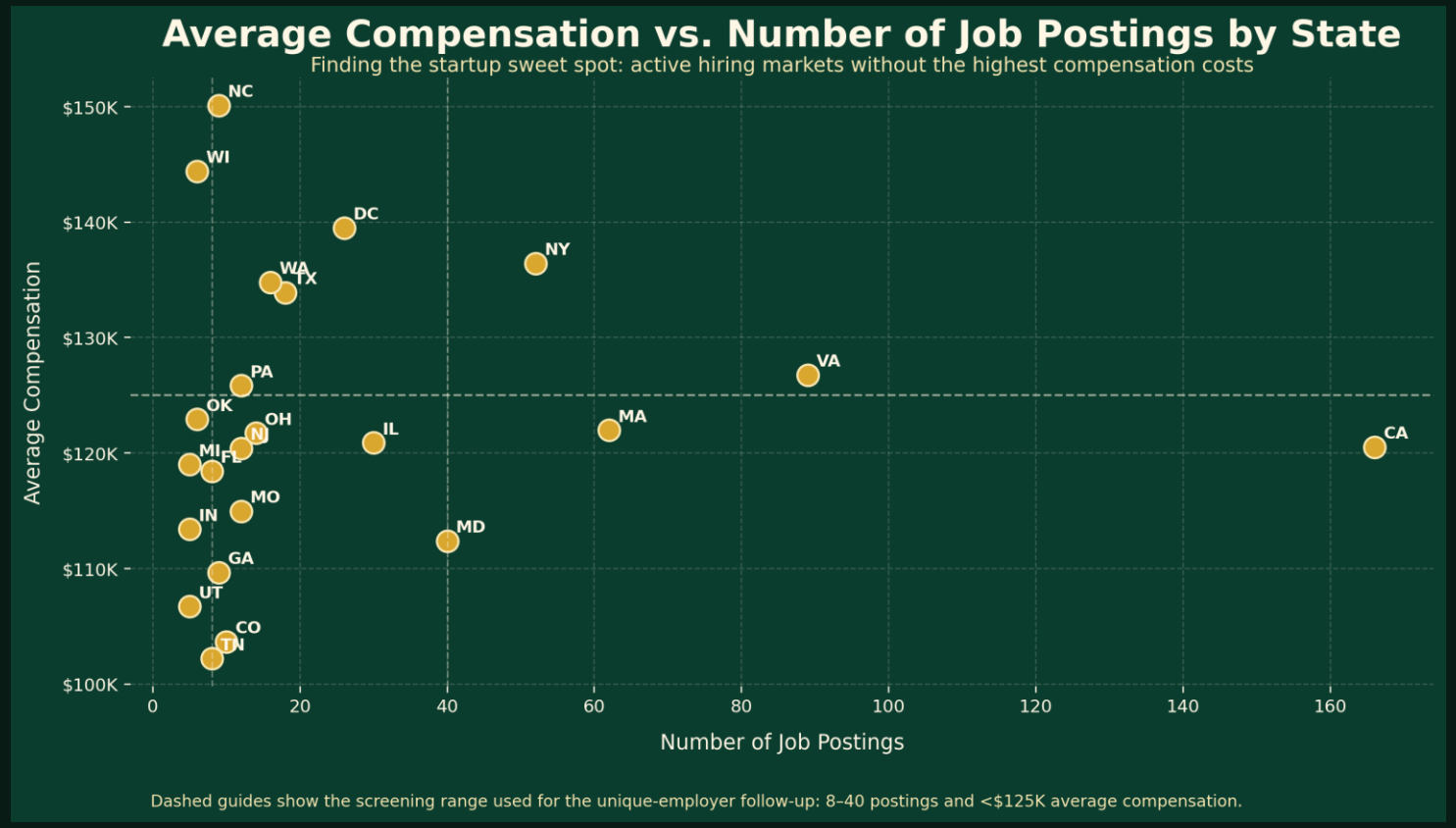
RECRUITING COMPETITION

The number of employers represented In each state gives us a proxy for how much competition w e may face when recruiting.

Our goal isn't to find the cheapest state – it's to find the best balance of affordability, market activity, and recruiting competition.

VISUALIZATION

Where Is Talent Affordable – but the Market Still Active?



WHAT WE FOUND

Several states offer compensation below $125K while maintaining meaningful job-posting activity.

STARTUP SCREEN

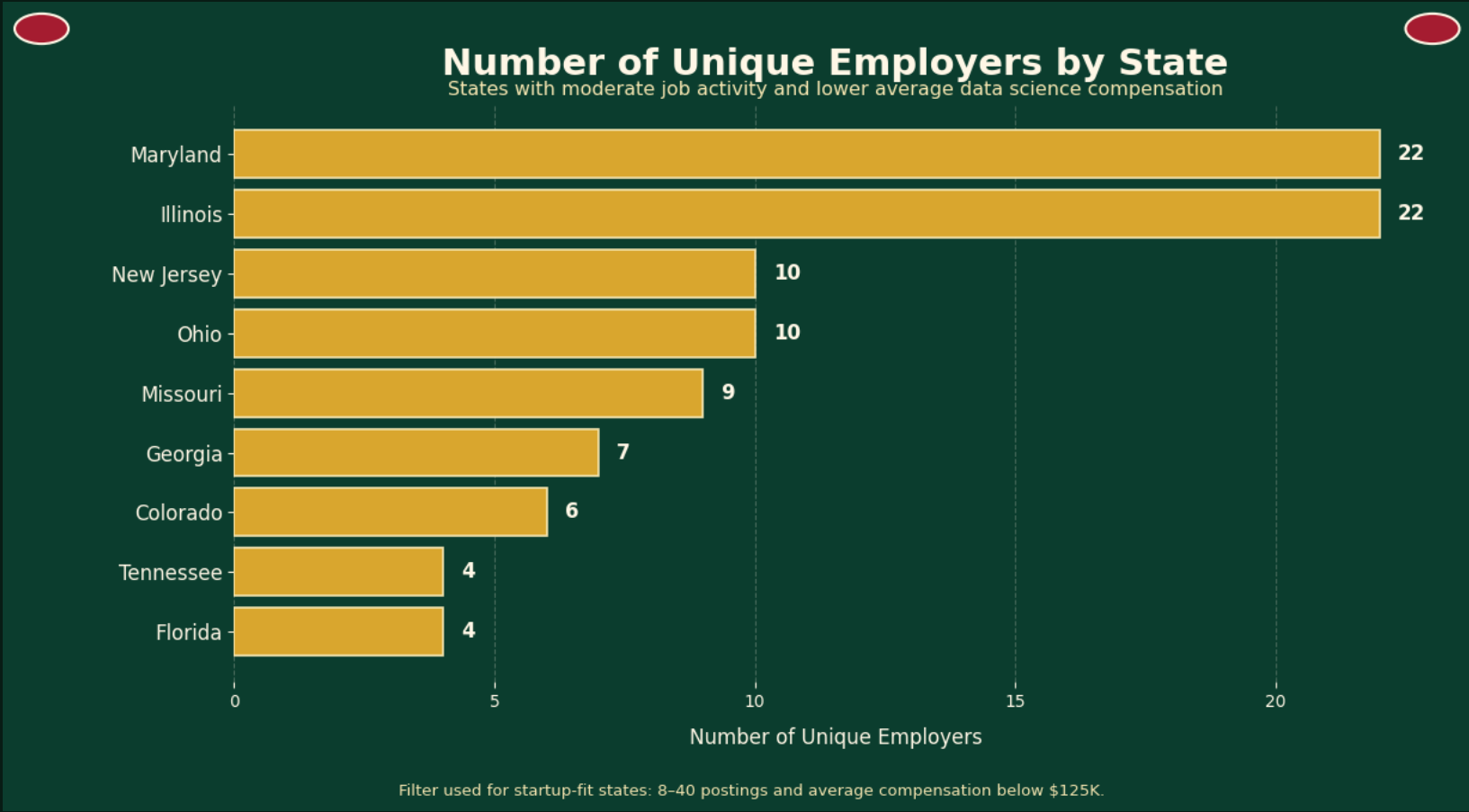
We narrowed the market to states with 8-40 postings and average compensation below $125K.

NEXT QUESTION

Which of these affordable, active markets also has manageable employer competition?

VISUALIZATION

Where is Employer Competition Most Manageable?



WHAT WE FOUND

Maryland and Illinois have the most employers represented, while Colorado remains comparatively low with 6.

THE TRADE-OFF

Lower employer competition alone doesn't make a market better; states with very few employers may also have lower posting activity.

Our Startup Sweet Spot: Colorado

$103.7K AVERAGE COMPENSATION

Among the lowest compensation levels in our screened markets.

10 JOB POSTINGS

Enough activity to pass our market-activity screening threshold.

6 UNIQUE EMPLOYERS

Relatively low employer presence compared with more crowded candidate markets.

Colorado doesn't lead every metric, but it offers one of the strongest balances across all three.

SUMMARY

Our Decision Framework

Clean data turns a vague hiring question into a defensible startup strategy.

1. KEY LESSONS

- Clean with the business question in mind
- Standardization prevents misleading comparisons
- Validate instead of making assumptions

2. IMPACT ON ANALYSIS

- Salary cleaning enabled accurate compensation comparisons across states.
- Location cleaning allowed job postings to be grouped and compared consistently by state.
- Company name cleaning allowed unique employers to be used as a proxy for recruiting competition

FINAL TAKEAWAY: Cleaning transformed inconsistent job-posting data into reliable measures of compensation, market activity, and employer competition.